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Dispersal and recruitment dynamics in the fleshy-fruited *Persoonia lanceolata* (Proteaceae)

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Abstract
Question: What is the role of dispersal, persistent soil seed banks and seedling recruitment in population persistence of fleshy-fruited obligate seeding plant species in fire-prone habitats?
Location: Southeastern Australia.
Methods: We used a long-term study of a shrubby, fleshy-fruited *Persoonia* species (*Proteaceae*) to examine (1) seed removal from beneath the canopy of adult plants; (2) seedling recruitment after fire; (3) the magnitude and location of the residual soil seed bank; and (4) the implications for fire management of obligate seeding species. We used demographic sampling techniques combined with Generalised Linear Modelling and regression to quantify population changes over time.
Results: Most of the mature fruits (90%) on the ground below the canopy of plants were removed by *Wallabia bicolor* (Swamp wallaby) with 88% of seeds extracted from *W. bicolor* scats viable and dormant. Wallabies play an important role in moving seeds away from parent plants. Their role in occasional long distance dispersal events remains unknown. We detected almost no seed predation *in situ* under canopies (< 1%). Seedling recruitment was cued to fire, with post-fire seedling densities 6-7 times pre-fire adult densities. After fire, a residual soil seed bank was present, as many seeds (77-100%) remained dormant and viable at a soil depth where successful future seedling emergence is possible (0-5 cm). Seedling survival was high (> 80%) with most mortality within 2 years of emergence. Plant growth averaged 17 cm per year. The primary juvenile period of plants was 7-8 years, within the period of likely return fire intervals in the study area. We predicted that the study population increased some five-fold after the wildfire at the site.
Conclusions: Residual soil seed banks are important, especially in species with long primary juvenile periods, to buffer the populations against the impact of a second fire occurring before the seed bank is replenished.

Keywords: Fire frequency; Macropod; Mammal; Obligate seeder; Soil seed bank.

Nomenclature: Harden, G.J. (1990-1992).

Introduction

Persistence of plant populations is influenced by a range of factors (Harper 1977). Ultimately, all plant populations rely on recruitment from seed to compensate for the death of individuals. For plants that occur in fire-prone habitats and are killed by fire (obligate seeders), recruitment is often linked to the passage of fire (Keeley 1991; Tyler 1995; Whelan 1995). The abundance and distribution of new recruits in such plant populations is controlled by a combination of factors. These include: (1) amount of seed produced (e.g. Enright & Lamont 1989); (2) seed dispersal to safe sites (Hughes & Westoby 1992; Lamont et al. 1993); (3) factors that govern the breaking of seed dormancy and promote seed germination (Auld & O'Connell 1991; Baskin & Baskin 1998; Keeley & Fotheringham 2000; Ooi et al. 2006); and (4) early seedling survival (Tyler 1995, 1996). As adults of obligate seeding species are killed by fire, sufficient time between fires is needed to replenish seed banks. Fire regimes that allow fires to repeatedly occur before seed banks are replenished may cause population declines or local extinction. Avoiding such fire regimes is necessary in order to manage fire-prone habitats for the conservation of biodiversity (Bradstock et al. 1998; Keeley & Fotheringham 2000; Gill 2001; Keith et al. 2002; Whelan 2002). An understanding of the dynamics of seed dispersal and factors influencing recruitment is necessary to determine how obligate seeding species respond to changes in fire regimes that result from direct human activities and climate change (Whelan 1986).

Seed dispersal of woody species in many fire-prone habitats can be limited, as indicated by the predominance of myrmecochory in SE and SW Australia (Berg 1975; Rice & Westoby 1981) and elsewhere (Gomez & Espadaler 1998). For such species, the capacity to re-occupy sites should a species be adversely affected by fire regimes is limited (Whelan 1986, 1995), especially where habitats are highly fragmented by urban and rural development. For some species, there is a greater

capacity for dispersal, through either wind dispersal of small seeds or dispersal by animal vectors. Fleshy-fruited species that may be moved by vertebrates are a small component of the heath and woodland flora of southeastern Australia (French & Westoby 1996) and other fire-prone floras (Bullock 1978; Milewski & Bond 1981; Keeley 1991). This may be the result of the low soil fertility that characterises the vegetation in these areas (Westoby et al. 1990). We estimate, based on seed morphology characteristics, that in the Sydney region, only some 9 % of species (in 57 genera) that occur in fire-prone heaths, woodlands and dry forest habitats have fleshy fruits that are likely to be dispersed by vertebrates. The most abundant groups are from two diverse families, *Persoonia* in the Proteaceae (29 taxa) and *Acrotriche*, *Astroloma, Brachyloma*, *Leucopogon, Lissanthe, Melichrus*, *Monotoca* and *Styphelia* in the *Ericaceae* (formerly included in the Epacridaceae, 43 taxa). The role of seed dispersal in maintaining populations or in moving seeds between populations in such taxa is poorly known.

Persoonia is an endemic Australian genus of ca. 100 species (Weston 2002), with two main centres of radiation, southeastern and southwestern Australia. Obligate seeding *Persoonia* spp. may have long primary juvenile periods, relative to the average inter-fire interval (Benson 1985; Auld unpubl.) and a number are threatened in the Sydney Basin of southeastern Australia (Rymer 2006). Such species are at risk of decline if fire frequencies are increased via arson, climate change or burning for protection of life and property.

In this paper we report on a long-term study of one obligate seeding shrub species, *Persoonia lanceolata* Andrews, that is common in eastern Australia and is a significant structural component of vegetation in heaths and woodlands. We ask:

1. Is seed removed or eaten from beneath the canopy of adult plants?
2. What agents are involved in seed removal or seed predation?
3. What is the magnitude of seedling recruitment after fire? Does this suggest population stability under the current fire regime?
4. What is the magnitude and location of the residual soil seed bank? How is this related to the pre-fire seed density? and
5. What are the implications for fire management of obligate seeding *Persoonia* spp.?

Methods

Study species and site

Persoonia lanceolata is a bushy shrub that grows to 3 m high. It is a fire sensitive obligate seeder, common in the understorey in heath and woodlands along the east coast and ranges of New South Wales, Australia (Weston 2002). Flowering occurs in summer (December to February) with mature fruits ripening in the following spring (September to October, Benson & McDougall 2000). The fruit of *Persoonia* species is a drupe (Weston 2002) of approximately 1300 mg (Westoby *et al*. 1990), with a pulpy flesh surrounding a hard woody pyrene. There is mostly one seed per pyrene in *P. lanceolata* (Westoby et al. 1990; pers. obs.). Seed dispersal is thought to be mostly by mammals and birds (Benson & McDougall 2000).

We examined a population of *P. lanceolata* at one site within Ku-ring-gai Chase National Park 20 km north of Sydney (33°53' E, 151°13' S) southeastern Australia. Annual precipitation for Sydney is around 1300 mm, while the average monthly maximum/minimum temperatures are 26/18 °C in summer and 16/8 °C in winter. The area of the site was ca. 2 ha and contained a dense mature population of *P. lanceolata* occurring in a periodically wet heathland. *Persoonia lanceolata* was one of the dominant shrubs along with *Angophora crassifolia*, *Eucalyptus stricta* and *Hakea teretifolia*. There was a range of smaller shrub species (0.5 - 2 m), particularly members of the *Proteaceae* and *Fabaceae* (common species include: *Banksia oblongifolia, Boronia pinnata, Dillwynia floribunda, Epacris pulchella, Grevillea speciosa, Hakea dactyloides, Leptospermum squarrosum, Mirbelia rubiifolia, Persoonia pinifolia, Pultenaea elliptica* and *Xanthorrhoea* spp.) and a ground cover of sedges, grasses, and herbs (common species include: *Actinotus minor, Anisopogon avenaceus, Bauera rubioides, Cyathochaeta diandra, Lindsaea linearis, Lomandra cylindrica, L. glauca, Mitrasacme polymorpha, Patersonia* sp., *Restio* spec.). The site was burnt by an extensive summer (January) wildfire in 1994 that burnt all the surrounding habitat. It had previously been burnt in a summer wildfire 15 years prior.

Seed dispersal and predation

Prior to the wildfire, we established plots under randomly selected mature *P. lanceolata* plants over two consecutive fruiting seasons. In the first season, we established five plots in a pilot experiment that ran for 12 days. In the second season, we established ten plots and followed the fates of seeds for 40 days. At each plot we established two treatments;

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(1) a cache of 10 mature fruits; and
(2) a cache of 10 mature fruits covered by a wire mesh cage (mesh 1 cm × 1 cm), cage 15 cm by 15 cm by 6 cm high. Each cage was placed over a petri dish and secured with wire stakes. These cages excluded vertebrates.

All caches were placed in 9 cm diameter petri dishes to avoid fruits being washed away in storms. The fate of seeds in caches was checked twice daily for 4 days then daily for 10 days and then 2-3 times per week. Searches were made around the caches for fruits or fruit/seed remains if seeds were removed from the caches. Where mammals were suspected of taking seeds (where pieces of seed testa were found or whole seeds were removed overnight) additional trials were established to identify the likely animals responsible. Adhesive tape was placed on the edges of petri dishes containing seeds and hair samples of the mammal species removing seeds were collected. These animal hairs were identified by B. Triggs. Field trapping using Elliott type A aluminium folding box traps was conducted to determine which small mammal species were present at the site.

A 2-factor GLIM was used to compare the proportion of caches where fruit were removed in the first 12 days in the caged and uncaged treatments, across two years (12 days being the length of time the experiment was run in the first season). A separate 1-factor GLIM examined the proportion of caches where fruit removal occurred across treatments at the end of the second season. In both cases a binomial error structure was used.

We also searched the surrounding habitat for any scats that contained pyrenes or fruits. We collected numerous scats and removed the pyrenes from them. For 10 replicate samples of 10 pyrenes from these scats we estimated the non-dormant seed fraction by placing replicate batches of seeds into 9-cm diameter petri dishes on Whatmans Grade 1 filter paper at room temperatures. Distilled water was added to ensure the filter paper remained moist and germination was followed for 2 months after which seeds were extracted from pyrenes and tested for viability using tetrazolium (Lakon 1949).

Seedling recruitment, survival and growth after fire

After the 1994 wildfire, we established 23 permanent 1-m^2 quadrats within the population using a randomised grid. Within each quadrat we scored the number of dead adults that were rooted in the quadrat and tagged new emerging seedlings of *P. lanceolata* for a period of 12 years. New seedlings (71) were identified by their distinctive cotyledons. At each sampling, we scored plant survivorship, assessed whether the plants had reached maturity (as indicated by the production of flowers and then fruits), and measured the height of all tagged *P. lanceolata* plants as an indicator of their growth to maturity. The time to first flowering was recorded for all seedlings and the magnitude of fruit production estimated in several flowering seasons by counting large (near mature) fruits on all tagged plants. The rate of growth of plants was examined using non-linear regression (power function) of plant height (lognormal transformed) against plant age. As plants were repeatedly sampled, we also carried out an Analysis of Covariance with previous sample plant height as a covariate. We also compared the pre-fire density of plants with the post-fire seedling density as an estimate of population turnover after the fire.

Residual soil seed bank

Within a randomly chosen subset (8) of the 23 available 1-m^2 quadrats described above to follow seedling emergence, survival and growth, we examined the residual soil seed bank. In 2003, 9 years after the 1994 fire (after seedlings had had their first flowering, but before the first post-fire seed fall) we excavated the soil in each of the 8 (1-m^2) plots. All the soil in each plot was excavated in two parts, i.e. 0 to 5 cm and 5-10 cm depths. Pyrenes were extracted by sieving. Samples were wet sieved by washing through firstly a 6.3 mm followed by a 4.2 mm sieve. The retained parts were allowed to dry and hand sorted. Pyrenes passed through the first sieve and were retained by the second. Intact pyrenes were opened to determine if the fruit contained a viable seed (filled, firm and white).

Results

Seed dispersal and predation

Seed removal outside cages after 12 days occurred in both sampled seasons (GLIM Δ dev = 14.1, 1 df, $P < 0.005$), with no interaction between years and treatment (GLIM Δ dev = 0.0, 1 df, $P > 0.05$) and no year effect (GLIM Δ dev = 0.1, 1 df, $P > 0.05$). Averaged across the two years some 17% of seed was removed (removal occurring in 53 % of seed caches) in 12 days (Fig. 1). At the end of the second sample season, where we followed the fate of seeds for 40 days, the treatment effect was again significant (GLIM Δ dev = 16.9, 1 df, $P < 0.005$), with an average of 90% of fruits removed in the open (removal occurring in 100 % of seed caches). The rate of seed removal outside the cages was slow but continued over the 40 day trial (Fig. 1). Inside cages, there was occasional consumption of the flesh around pyrenes, and occasionally whole fruits were removed. Outside of cages most losses were fruit removals, although in a very few cases the pyrene was removed and the flesh remained.

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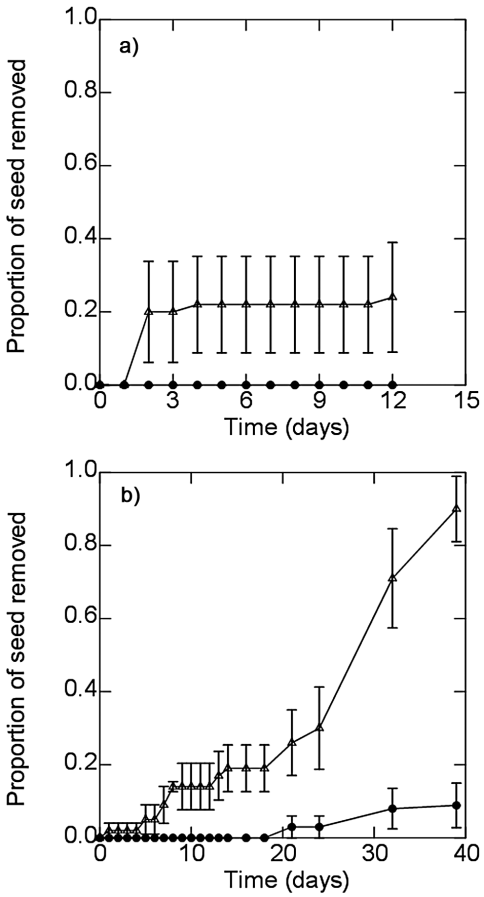


Fig. 1. Seed removal rates over two annual fruit-crops in *Persoonia lanceolata*, (**a**) year 1; (**b**) year 2. Treatments are: caged for vertebrate exclusion (filled circles); uncaged (open triangles). Error bars are 1 SE.

Only two fruits in the experiment in the first year were observed to have been eaten *in situ*. Where whole seed removals occurred, two lines of evidence suggested that the fruits were eaten by the macropod *Wallabia bicolor* (swamp wallaby). Hair samples from around batches of seeds were identified as this species and scats of the species frequently contained *Persoonia* pyrenes. Field trapping of small mammals using Elliott Traps showed that only *Rattus fuscipes* and *Antechinus* sp. (a marsupial mouse) were commonly trapped at the site. Neither species is known to remove large fleshy fruits.

For the pyrenes extracted from *Wallabia bicolor* (Swamp Wallaby) scats we found high seed viability (88 ± 3%), but all viable seeds were still dormant.

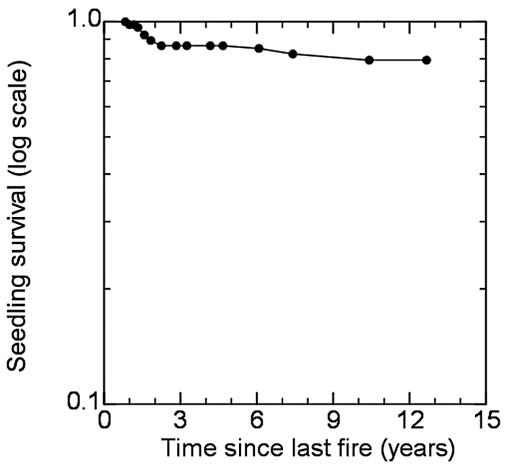


Fig. 2. Survival of seedlings of *Persoonia lanceolata* over time.

Seedling recruitment, survival and growth after fire

Most seedlings emerged in the first 10 months after the 1994 fire (the time of our first sample). There was only limited emergence after this (Table 1), but occasional seedlings appeared until 6 years post-fire. This subsequent emergence was not confined to any particular season (summer 11, autumn 6, winter 3 seedlings, respectively), and declined in magnitude over time. Seedling survival was high with most observed mortality in the first 1-2 years after seedlings were tagged (Fig. 2). Our sampling did not allow us to detect mortality in the first 10 months post-fire. The post-fire seedling density was 6-7 times greater than the pre-fire adult density (0.5 vs. 3.3/m^2).

Seedlings reached a height of ca. 1 m (range 0.10 - 1.7m) at 6-7 years old, growing at an average rate of 0.17 m per year (Fig. 3). Age was a good predictor of plant

Table 1. Fecundity data for maturing juveniles of *Persoonia lanceolata*.

Plant age (years)	N[1]	Cumulative proportion of seedlings	Plants that have fruited (%)	Annual fruit production per plant (Mean ± standard error)
0	0			
1	59	0.83	0	0
2	70	0.99	0	0
3	70	0.99	0	0
4	70	0.99	0	0
5	70	0.99	0	0
6	60	1.0	13	0.8 ± 0.3
7	57		35	30.3 ± 10.6
10	25[2]		72	N/A
12	25[2]		92	251.0 ± 66.5

[1]Number of tagged plants in sample;
[2]Number reduced as some plots were excavated for residual seed bank sampling.

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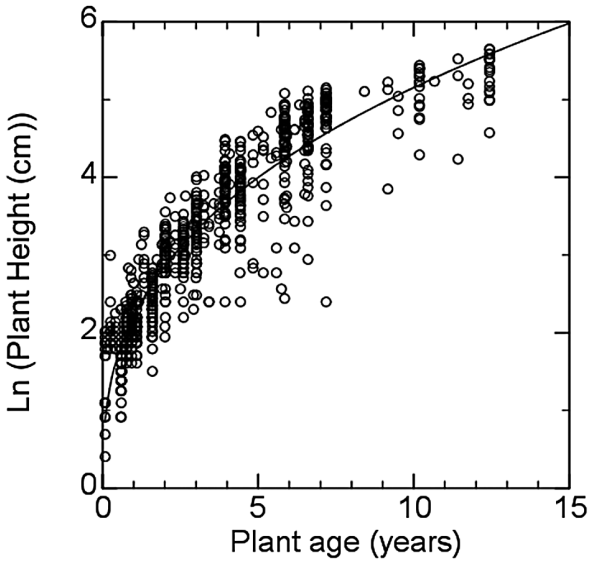


Fig. 3. Plant height against age for seedlings of *Persoonia lanceolata*. The line refers to the power function: Ln(plant height) = 2.215*(plant age)$^{0.367}$; $R^2 = 0.87$).

height with a power function of log height accounting for some 87% of the variation in mean plant height. The Analysis of Covariance showed a significant effect of both plant age ($P < 0.001$) and previous plant height ($P < 0.001$), suggesting that although a number of smaller plants persist for a while (see Fig. 3), the height of a plant at a particular age was not simply dependent on its previous height.

No plants flowered until 6 years after the fire. At this time only 13% of plants were mature and resultant fruit production was very low (Table 1). The proportion of plants that reproduced increased over time up to 92% 12 years after the fire. The magnitude of fruit production also increased markedly from the first to the second fruiting season and in subsequent seasons (Table 1).

Residual soil seed bank

Residual seeds in the soil were found in all sampled quadrats, independent of whether or not above ground seedlings were present at the time of sampling (Fig. 4). The density of seeds in the soil (21.2 ± 5.1 seeds/m^2) was some 6-7 times greater than the density of seedlings that emerged after fire. On a 1-m^2 plot basis, we estimate that between 72-100% of the available soil seed bank did not germinate after the wildfire. This may be an over-estimate, as some seeds that germinated may not have been observed as seedlings at our first census 10 months after the fire. Most of the residual soil seed bank was in the top 5 cm of soil (90%).

Discussion

Fire plays an important part in the life cycle of many plant species, including woody *Persoonia* species. Like many other obligate seeders, for the common *P. lanceolata*, fire caused population-scale adult mortality and promoted the recruitment of new plants. Population persistence in this obligate seeding species will be influenced by fire frequency, fire patchiness and the dispersal capability of seeds. After a single fire, most seedling recruitment was confined to the first 1-2 years, suggesting that seeds of this species respond to fire cues and/or that the post-fire environment provides favourable conditions for seedling establishment. For obligate seeding shrubs in general, recruitment is often linked to the first 12 months after fire, e.g. southeastern Australia (Purdie 1977; Auld & Tozer 1995; Benwell 1998); Spanish matorral (Fernández-Santos & Gómez-Guitérrez 1994); Florida sand pine scrub (Carrington 1999); Californian chaparral and sage brush (Keeley et al. 2006). However, there was no seasonal component observed in seedling emergence after the initial flush (in comparison to Auld et al. 2000; Ooi et al. 2004). Some continued seedling recruitment occurred for up to 6 years post-fire.

The number of seedlings recruited post-fire exceeded pre-fire adult numbers by 6-7 times. The likely change in population abundance after the wildfire can be compared using an estimate of pre-fire adults to post-fire seedlings, and incorporating estimated subsequent seedling mortality levels. This included mortality prior to plants reaching maturity (Fig. 2, ca. 14 (± 4)% after 7.5 years) and an estimate of the likely subsequent adult mortality in the following 7.5 years (to allow equivalent populations estimates for a 15-year old stand). We assumed mortality from 7.5-15 years post-fire matched that of the 2-12 year old plants in the study (Fig. 2). Based on these estimates, we predict that the population density 15 years after the

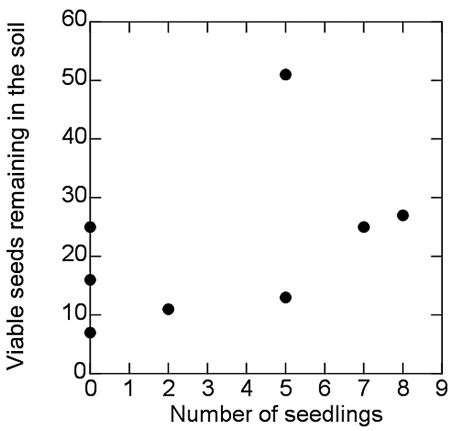


Fig. 4. Magnitude of the residual soil seed bank in *Persoonia lanceolata* in 1-m^2 quadrats.

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1994 wildfire would be some 5 × that of the pre-fire density. Few comparable examples of quantitative estimates of population change in response to fire are available. Keeley et al. (2006) observed an increase in abundance in obligate seeding shrubs in the first five years post-fire in southern California, whilst Auld & Scott (2004) showed that the ratio of pre vs. post-fire plant abundance in obligate seeders was dependent upon the time since the previous fire (i.e. pre-fire plant ages). This general lack of data in fire-prone habitats limits quantitative comparison of the impact of varying fire regimes on persistence of obligate seeding plants.

Survival of new seedlings of *P. lanceolata* is of a Deevey Type III (*sensu* Harper 1977), with highest mortality when plants are young and small, and reduced mortality once the seedlings are established and grow. This matches the pattern for seedlings in other fire-prone habitats (Fernández-Santos & Gómez-Guitérrez 1994; Whelan 1995). Plant growth is relatively slow compared to co-habiting obligate seeding shrubs and in this study, flowering did not occur until 6 years after the fire. As it takes over 6 months for fruits to mature and as the first flowering effectively produced almost no seed, we consider the first effective fruit set was some 7-8 years after the fire. This relatively long primary juvenile period has been noted at other sites for this species (Benson & McDougall 2000; Auld unpubl.), although it has also been observed to be slightly shorter (Benson & McDougall 2000). It is also typical of other co-habiting, obligate seeding *Persoonia* spp. such as *P. pinifolia* and *P. isophylla* (Benson & McDougall 2000; Auld unpubl.). Fires may readily re-occur in the habitat of *P. lanceolata* every 5-10 years, with most (75%) areas in nearby comparable vegetation having a current fire return interval of 7-17 years (Bradstock & Kenny 2003). Consequently, the species may be at risk of population declines unless either a significant proportion of the soil seed bank remains available after a fire (as a residual seed bank), or seeds can readily disperse back into the population from unburnt populations.

We found considerable residual seed at our study site and much of this lay in the top 5 cm of the soil, the area most likely to be influenced by fire-related germination cues such as heat (Bradstock & Auld 1995), and possibly smoke. These buried seeds are unlikely to be killed by the lethal heating effects of a fire. Seeds in the top 5 cm will also be least affected by limitations to successful emergence due to depth constraints (Bond et al. 1999). Residual seed values varied, but all greatly exceeded those of some other obligate seeders that co-occur with *P. lanceolata* (Auld & Denham 2006). However, even the residual soil seed bank detected in this study does not ensure there will be population level replacement should a fire occur before there has been replenishment of the soil seed bank. The magnitude of any germination event will be governed by factors relating to the fire itself (heat and smoke output) and post-fire rainfall levels.

Removal of fruits from below parent shrubs by Swamp Wallabies was common. In contrast, removal by invertebrates was uncommon, as observed for this species at other sites by French & Westoby (1996). Very few seeds were consumed below canopies. In contrast to some other members of the *Proteaceae* such as *Grevillea* (Auld & Denham 1999), seed removal does not seem to be a mechanism for seed escape from seed predators. Rather, seed removal by Swamp Wallabies would likely reduce future seedling densities adjacent to pre-fire parent plants. Seeds that were removed passed though the gut of animals in a viable and dormant state. Hence, Swamp Wallabies are likely to be an effective dispersal agent for *Persoonia* spp., in agreement with Rymer (2006). Most seeds are likely to be deposited within a population, as evidenced by numerous scats containing seeds in the population. The question remains as to whether these dispersal agents have the ability to move seeds between populations, especially in order to prevent local population extinctions under a regime of too frequent fire. It has been suggested that vertebrate dispersal of *Persoonia* seeds may be a mechanism for gene flow between populations (Rymer & Ayre 2006) or for re-establishment of populations after local extinction (French & Westoby 1996).

The long-term nature of this study allowed us to quantify population processes and changes over time. Such studies are a trade off between parameter estimation and predictability across the landscape (Silvertown & Lovett Doust 1983; Whelan 1995). The utility of these data for making predictions about other populations of *Persoonia lanceolata* remains to be investigated.

Conclusions

The observed maintenance of a large post-fire residual soil seed bank was potentially an effective means of buffering the population from local extinction if fires occasionally re-occur at short intervals. A long-term high frequency fire regime that prevents the replenishment of the soil seed bank is likely to lead to population declines and possible local extinction. We suggest that a sound understanding of how fire impacts on obligate seeding species, such as *Persoonia,* will be best developed by a combination of long-term studies at particular sites combined with detailed multi-site or multi-species comparisons of key demographic attributes, e.g. seed germination, seed dispersal patterns, residual soil seed banks.

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